

$$\left. \begin{aligned} y &= \frac{B}{A} u \\ y &= \frac{KB}{PA+QB} \end{aligned} \right\} u = \frac{R}{P} w - \frac{Q}{P} y \Rightarrow Pu = Rw - Qy$$

$$\Rightarrow p_0 u(k) + p_1 u(k-1) = R w(k) - q_0 y(k) - q_1 y(k-1)$$

$$u(k) = \frac{R w(k) - \sum q_i y(k-i) - \sum p_j y(k-j)}{p_0}$$